SILTHAUL · PROBLEM STATEMENT

SiltHaul problem statement

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 2 update; first co-design candidate named (SLH-DDR-001); questions for the first trials
0.1	2026-09-30	Amish Chadha	Initial scaffold

SiltHaul problem statement

Flood mud is heavy, wet and contaminated, and it sits in the places hardest to reach with machines: ground-floor rooms, narrow lanes and drains. Clearing it is left to people with shovels and buckets.

The problem

Mud has to be dug, lifted, carried out and dumped, often through a single door. OSHA lists back, knee and shoulder injuries from manual lifting among the main risks for flood clean-up workers and recommends teams for heavy items (OSHA). The longer people spend in the mud, the longer they are exposed to contaminated water; after the Kerala floods, officials linked leptospirosis risk to contact with water contaminated by animal urine and urged clean-up volunteers to take preventive medicine (Mongabay, 2018).

Mechanical options exist but do not fit. Mines use powered double-drum scraper hoists, or slushers, that drag a scraper box on a pull rope and return it on a tail rope (Mining History Association); a recent Sulzer design uses electric motors inside the drums and has ceased without grant (WO2017216731A1). These are heavy, powered and built for mine drives. Excavators and loaders cannot enter homes or narrow lanes. There is no portable, hand-powered slusher for flood recovery.

Users and context

Table 1. Users and needs.

User type	Goals and needs	Typical use scenario	Pain points and challenges
Householders and volunteers	Get mud out of rooms faster and with less lifting and less time in the mud	Ground floors of homes, shops and places of worship after a flood	Heavy buckets, back and shoulder strain, long contact with contaminated water and silt
Relief NGOs and community groups	A kit that is cheap, portable, needs no fuel and can be used by untrained volunteers after a short briefing	Many sites, poor road access, limited power	Equipment that needs power, special skills or a truck; volunteers turn over daily
Municipal drain and sanitation crews	Clear silted drains and lanes where machines cannot reach	Narrow streets, open drains, culverts	Excavators and loaders cannot enter; manual clearing is slow
Local fabricators	Drawings using stock steel, rope and pulleys	Workshops near flood-prone areas	No drawings for a hand slusher; unsure of weld and fit requirements

Stakeholders

Table 2. Stakeholders.

Stakeholder group	Role or interest	Engagement needed
Disaster management authorities and civil defence volunteers	Run flood clean-up and set its safety rules	Co-design partner candidate: trial rooms, crews, safety review (not agreed)
Local health authorities	Advise on contamination, protective clothing and hygiene	Review of the safety notes and the briefing card
Building owners and landlords	Own the slab the tail block is drilled into	Permission to drill; rule that walls are never anchors
Rope, rigging and bearing suppliers	Supply rated sling, shackles, rope, chain and bearings	Quotes and rated parts at TRL 4
Standards and regulatory advisers	Machinery and lifting-gear rules of the country of use	Review of SLH-REG-001 items marked "to assess"

Operating environment

- Wet silt and mud of varying depth, from a thin layer to deep deposits; mixed with debris, rubbish and sometimes sewage.
- Ground-floor rooms with doorways about 0.7 to 0.9 m (28 to 35 in) wide (estimate), steps and thresholds.
- Lanes and open drains in dense neighbourhoods.
- Walls that may be weakened and wet after the flood.
- Hot, humid conditions, or cold and wet in temperate regions; no power.

Constraints

- Prototype parts budget: USD 2,000 or less.
- Hand powered only.
- Fits through a standard doorway and into a car or small pickup.
- Anchors must not rely on flood-damaged walls; the tail sheave needs its own anchoring option.
- Buildable with hand tools and a small welder from stock steel, rope and commercial pulleys.
- Open design: hardware under CERN-OHL-S-2.0, any calculators under MIT.

Out of scope

- Pumping standing water.
- Removing large debris such as furniture, vehicles or fallen structures.
- Motorised versions.
- Disposal and treatment of the removed mud.

Prior work

Prior work	What it does	Gap for these users	Source
Mine slusher (tugger-scraper)	Powered two-drum hoist drags a scraper along a mine floor and returns it by tail rope through a sheave	Compressed-air or electric, heavy, built for mines	link
US2588657A, slusher bucket (expired)	Scraper bucket worked by lead, return and tail ropes on a portable hoist	Mining equipment for a powered hoist	link
US3532170A, slusher scraper bucket and blade assembly (expired)	Scraper bucket and blade design for slushers	Mining equipment for a powered hoist	link
WO2017216731A1, scraper winch (Sulzer, ceased)	Double-drum scraper winch with electric motors inside the drums and remote control	Powered mine winch; no hand operation	link

Co-design

A disaster relief NGO, a municipal disaster management unit or a volunteer fire service with regular flood clean-up experience, who can supply realistic mud and building conditions and test the kit with volunteers. First candidate to approach (not agreed): the Kerala State Disaster Management Authority and its civil defence volunteers, since Kerala's 2018 clean-up is the case that sparked the idea (SLH-DDR-001).

Safety

> **Safety:** Any tool that moves mud by rope puts volunteers near ropes and anchors under tension, in buildings that may be damaged and in mud that may carry sewage and disease. The design answers are in SLH-PRC-001, Safety; the clean-up hazards themselves (structure, electricity, contamination) stay with the users and local health advice.

Questions for the first trials

These are settled by trials (TRL 4), not by design decisions:

- How well does the box fill and drag in different muds, from liquid slurry to stiff clay?
- How often is the floor slab sound enough for the tail block's drilled anchors (the only tail anchor used, SLH-DDR-001)?
- Can one capstan serve several rooms in turn without re-anchoring outside?
- How does the rope get past steps and turns in the building? The doorway ramp covers thresholds up to 110 mm; turns are out of scope for the prototype.
- Is a smaller drain version, pulled along an open channel, worth a separate variant?